In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - IV

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 186**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 06)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (07 to 14) contains 8 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer and **– 2 mark** for wrong answer.

Section-A (15 – 18) contains 4 Paragraph Type questions: Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct. Each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

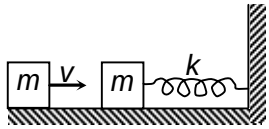
SECTION – A

Straight Objective Type

This section contains **6 multiple choice questions** numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. A particle is projected from a point *O* with velocity *u* in a direction making an angle α with the horizontal. At *P*, it is moving at right angles to its initial direction of projection. Its speed at *P* is
(A) $u \tan \alpha$ (B) $u \cot \alpha$
(C) $u \operatorname{cosec} \alpha$ (D) $u \sec \alpha$

2. A block of mass *m* moving with velocity *v* strikes elastically another identical mass connected to a spring as shown in figure. The maximum compression produced in the spring (Assume surfaces to be smooth) is



(A) $\sqrt{\frac{mv^2}{2k}}$ (B) $\sqrt{\frac{mv^2}{k}}$
(C) $\sqrt{\frac{mv^2}{3k}}$ (D) none of these

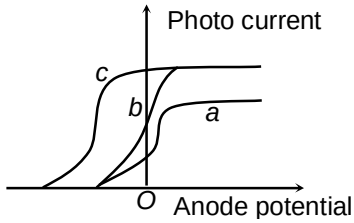
3. A block of mass 20 kg and volume 10^3 cm^3 is suspended vertically from a ceiling by a wire. The linear mass density of the wire is 5×10^{-3} kg/m and its length is 50 cm. The wire is vibrating in its fundamental mode and producing beats with a tuning fork of frequency 190 Hz. When the block is just completely immersed in a liquid and vibrated in its fundamental mode, it produces the same number of beats with tuning fork as earlier. Density of the liquid is ($g = 10$ m/s^2)
(A) 3.8 gm/cm^3 (B) 7.6 gm/ cm^3
(C) 1.9 gm/ cm^3 (D) 5.0 gm/ cm^3

4. A vertical cylinder with a massless piston filled with one mole of an ideal gas. The piston can move freely without friction. The piston is slowly raised so that the gas expands isothermally at temperature 300 K. The amount of work done in increasing the volume by two times is
($R = \frac{25}{3}$ J/mol/K , $\log_e 2 = 0.7$)
(A) 1750 J (B) 2500 J
(C) 750 J (D) 4250 J

Space for rough work

5. In Young's double slit experiment, double slit of separation 0.1 cm is illuminated by white light. A coloured interference pattern is formed on a screen 100 cm away. If a pin hole is located on this screen at a distance of 2 mm from the central fringe, the wavelength in the visible spectrum which will be absent in the light transmitted through the pin-hole are
(A) 5714 Å and 4444 Å (B) 6000 Å and 5000 Å
(C) 5500 Å and 4500 Å (D) 5200 Å and 4200 Å

6. The figure shows the variation of photo current with anode potential for a photo-sensitive surface for three different radiations. Let I_a, I_b and I_c be the intensities and f_a, f_b and f_c be the frequencies for the curves *a*, *b* and *c* respectively;



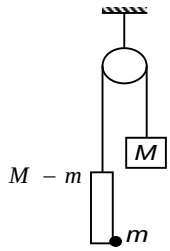
(A) $f_a = f_b$ and $I_a \neq I_b$
(B) $f_a = f_c$ and $I_a = I_C$
(C) $f_a = f_b$ and $I_a = I_b$
(D) $f_b = f_c$ and $I_b = I_c$

Multiple Correct Choice Type

This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

7. A particle is acted upon by a force of constant magnitude which is always perpendicular to the velocity of the particle. The motion of the particle takes place in a plane. It follows that
(A) its velocity is constant (B) its acceleration is constant
(C) its kinetic energy is constant (D) it moves in a circular path

8. A rod of mass '$M - m$' carries an insect of mass '*m*' at its bottom end and its top end is connected with a string which passes over a smooth pulley and the other end of the string is connected to a counter mass *M*. Initially the insect is at rest. Choose the correct option(s).



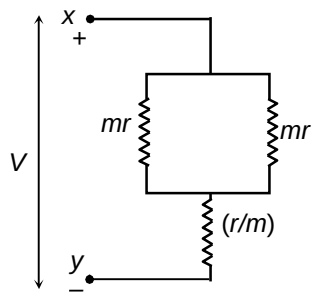
(A) As insect starts moving up relative to rod, the acceleration of centre of mass of the system (insect + rod + counter mass) becomes non-zero
(B) As insect starts moving up relative to rod, tension in the string remains constant and is equal to *Mg*.
(C) As insect starts moving up relative to rod, the tension in the string becomes more than *Mg*.
(D) Acceleration of centre of mass of the system (insect + rod +counter mass) is zero when insect moves with constant velocity.

Space for rough work

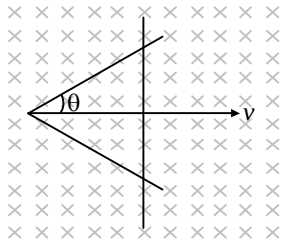
9. A particle of mass *m* moves in a straight line. If *v* is the velocity at a distance *x* from a fixed point on the line and $v^2 = a - bx^2$, where *a* and *b* are constant, then
(A) The motion continues along the positive *x*-direction only
(B) The motion is simple harmonic
(C) The particle oscillates with a frequency equal to $\frac{\sqrt{b}}{2\pi}$
(D) The total energy of the particle is *ma*

10. 10 gms of ice at 0^0C is mixed with 5 gms of steam at 100^0C. If latent heat of fusion of ice is 80 cal/gm and latent heat of vaporization is 540 cal/gm. Then at thermal equilibrium
(A) temperature of mixture is 0^0C
(B) temperature of mixture is 100^0C
(C) mixture contains 13.33 gms of water and 1.67 gms of steam
(D) mixture contains 5.3 gms of ice and 9.7 gms of water

11. In the given circuit the value of *m* is varying. The correct statements about the circuit are
(A) The condition for maximum current flowing from *x* is *m* =2.
(B) The maximum current is $\frac{V}{\sqrt{2}r}$.
(C) The condition for maximum current flowing from *x* is $m = \sqrt{2}$.
(D) The maximum current is $\frac{3V}{2r}$.

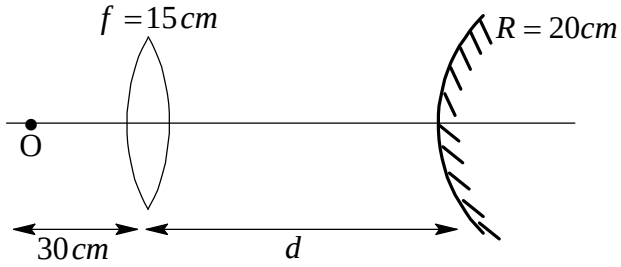


12. Two straight conducting rails form a right angle where their ends are joined. A conducting bar in contact with the rails start at the vertex at $t = 0$ and moves with a constant velocity *v* along them as shown. A magnetic field *B* is directed into the page. The induced emf in the circuit at any time *t* is proportional to
(A) t^0 (B) t
(C) v (D) v^2



Space for rough work

13. The value of d for which the final image coinside with the object is/are:



(A) $10\,cm$ (B) $20\,cm$
(C) $30\,cm$ (D) $40\,cm$

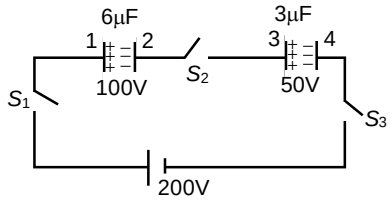
14. Hydrogen gas absorbs radiations of wavelength λ_0 and consequently emit radiations of 6 different wavelengths of which two wavelengths are shorter than λ_0. Choose the correct statement(s).
(A) The final excited state of the atom is $n = 4$
(B) The initial state of the atom may be n = 2
(C) The initial state of the atom may be n = 3
(D) There are three transitions belonging to Lyman series.

Comprehension Type

This section contains **2 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Questions 15 & 16

Two capacitors of capacity 6 μF and 3μF are charged to 100V and 50V separately and connected as shown in figure. Now all the three switches S_1, S_2 and S_3 are closed.



15. Charges on the 6μF capacitor in steady state will be
(A) 400 μC (B) 700 μC
(C) 800 μC (D) 250 μC

Space for rough work

16. Suppose q_1, q_2 and q_3 be the magnitudes of charges flown from switches S_1, S_2 and S_3 after they are closed. Then

(A) $q_1 = q_3$ and $q_2 = 0$ (B) $q_1 = q_3 = \frac{q_2}{2}$

(C) $q_1 = q_3 = 2q_2$ (D) $q_1 = q_2 = q_3$

Paragraph for Questions 17 & 18

In physics laboratory, Peter is trying to calculate the focal lengths of a convex lens. He measures the distance between a screen and a light source lined up on the optical bench to be 120 cm. When Peter shifts the convex lens along the axis of optical bench, sharp images of the source is obtained at two lens positions. He also measures the ratio of these two magnifications to be 1 : 9.

17. The focal length of the convex lens measured by Peter is
(A) 22.5 cm (B) 30 cm
(C) 45 cm (D) none of these

18. Which image, as seen by Peter is brighter?
(A) smaller (B) bigger
(C) both are equally bright (D) cannot be judged

Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **6 multiple choice questions** numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. In a crystal structure of an element, atoms are present in fcc arrangement as well as in tetrahedral voids. Which of the following relation between the atomic radius and the edge length is correct?

   (A) $\sqrt{2}a = 4r$ (B) $\sqrt{3}a = 8r$

   (C) $\sqrt{3}a = 4r$ (D) $a = 2r$

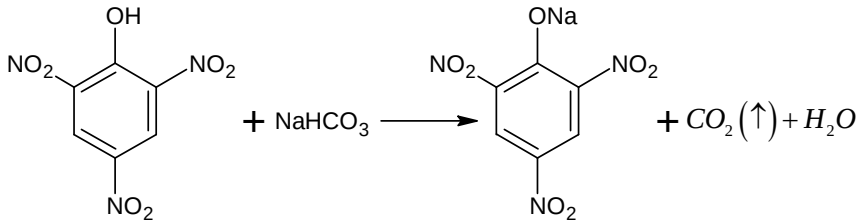
2. Which of the following does not contain transition metal?

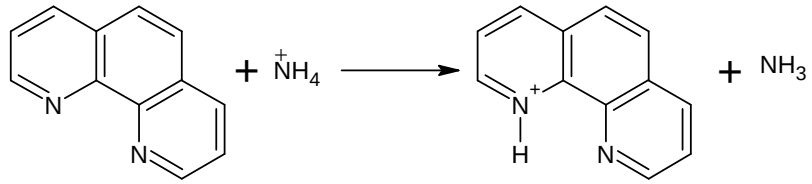
   (A) Haemoglobin (B) Vitamin B_{12}

   (C) Cis – platin (D) Chlorophyll

3. Which one of the following reactions is not correct?

   (A) $CH_3COOK + CF_3COOH \longrightarrow CH_3COOH + CF_3COOK$

   (B) 

   (C) 

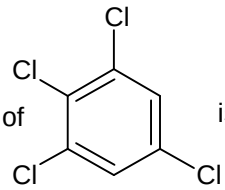
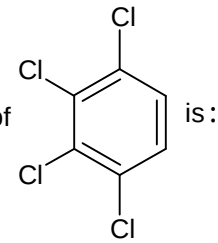
   (D) $CH_3CH_2OH + NaHCO_3 \longrightarrow CH_3CH_3ONa + H_2O + CO_2(\uparrow)$

Space for rough work

4. The stopping potentials in an experiment of photoelectric effect are V_1 and V_2, with incident light of wavelength λ and $\frac{\lambda}{2}$ respectively. The correct relation between V_1 and V_2 may be:

(A) $V_2 > 2V_1 > V_1$ (B) $V_1 < V_2 < 2V_1$

(C) $V_2 = 2V_1$ (D) $V_2 < 2V_1$

5. The dipole moment of  is $1.5\,D$. The dipole moment of  is:

(A) zero (B) $1.5\,D$

(C) $2.6\,D$ (D) $2.12\,D$

6. Which of the following is not the compound of aluminium?

(A) anhydrite (B) bauxite

(C) corundum (D) Diaspore

Multiple Correct Choice Type

This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

7. Which of the following relation is/are correct about $NaCl$ structure (a is the edge length and r is the radius of atom)?

(A) $(a) > \left(2 \times r_{Cl^-}\right)$

(B) $\left(\sqrt{2} \times a\right) > \left(4 \times r_{Cl^-}\right)$

(C) $(a) = 2\left(r_{Cl^-} + r_{Na^+}\right)$

(D) Cl^- ions have expended fcc arrangement

Space for rough work

8. When 5 mL of a 1.0 M HCl solution is mixed with 5 mL of a $0.1M$ $NaOH$ solution, temperature of solution increases by $t°C$. Which of the following(s) can be predicted from this observation:

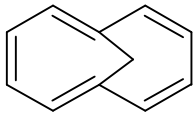
(A) If $5mL$ of $0.1M$ HCl is mixed with 5 mL 0.1 M NH_3 solution, the temperature rise will be less than $t°C$

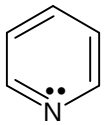
(B) If 5 mL 0.1 M CH_3COOH is mixed with 5 mL 0.1 M $NaOH$, the temperature rise will be less than $t°C$

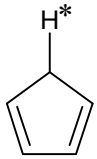
(C) If 10 mL of 1.0 M HCl is mixed with 10 mL of 0.1 M NaOH, the temperature rise will be $2t°C$.

(D) If 10 mL of 0.1 M HCl is mixed with 10 mL of 0.1 M NaCl, the temperature rise will be $t°C$

9. Identify correct statement/statements from the following options:

(A)  has aromatic system

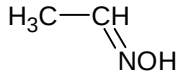
(B)  has cyclic $6\pi e^-$ system

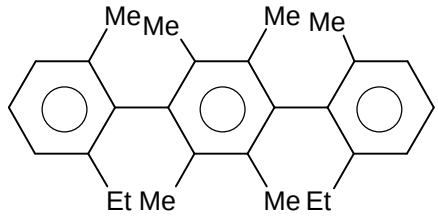
(C) Marked hydrogen in  is more acidic then marked hydrogen in 

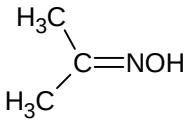
(D)  is unstable at room temperature

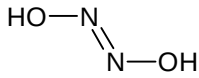
Space for rough work

10. Which will show geometrical isomerism?

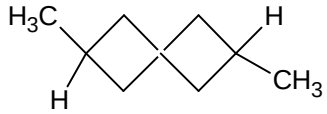
(A) 

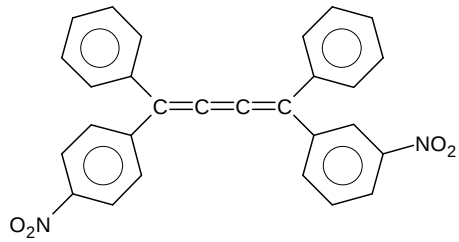
(B) 

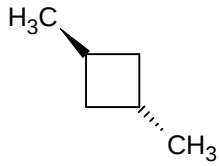
(C) 

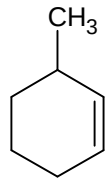
(D) 

11. Identify the chiral molecule/molecules:

(A) 

(B) 

(C) 

(D) 

12. Which of the following is/are can produce H_2O_2 on hydrolysis?

(A) BaO_2 (B) PbO_2

(C) MnO_2 (D) Na_2O_2

Space for rough work

13. The correct statement/statements is/are:
(A) The distance between two octahedral voids of a cubic close packing of atoms with radius r is 2r
(B) The distance between two tetrahedral voids of a cubic close packing of atoms with radius r is $\sqrt{6} \times r$ at the body diagonal axes
(C) The distance between tetrahedral and octahedral void is $\sqrt{\frac{3}{2}} \times r$, where is the radius of atom forming the close packing
(D) The distance between the surface of atoms fcc arrangement along an edge is $2\left(\sqrt{2}-1\right)r$

14. Identify the correct relationship:
(A) $\left[Cu\left(NH_3\right)_4\right]^{2+}$ with Hybridization of $Cu = dsp^2$
(B) $Fe\left(CO\right)_x$ with $x = 5$
(C) Zn^{2+} Complexes with sp^3d^2 hybridization are diamagnetic in nature
(D) Cr^{3+} Complexes with Paramagnetic nature

Comprehension Type

This section contains **2 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Question Nos. 15 to 16

Assume that water vapour behaves as an ideal gas and the volume occupied by the liquid water is negligible compared to the volume of the container. A sample of liquid water of mass 3 g is injected into an evacuated 76 L flask maintained at 320 K. At this temperature the vapour pressure of water is 32 mm of Hg:
Now give the answers to following questions:

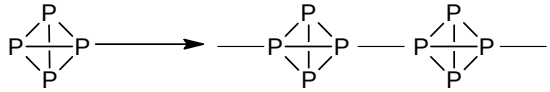
15. What % of the water will be vaporised, when the system comes to equilibrium:
(A) 18% (B) 73%
(C) 30% (D) 64%

16. What should be the minimum volume of the flask if no liquid water is to be present at equilibrium?
(A) $174L$ (B) $150L$
(C) $104L$ (D) $126L$

Space for rough work

Paragraph for Question Nos. 17 to 18

White phosphorus is a tetra atomic solid $\left[P_4(s)\right]$ at room temperature and on strong heating in absence of oxygen, it polymerizes into red phosphorus as:



ΔH=-104KJ/mol of P_4

White(g) red(g)

The enthalpy of sublimation $\left[P_4(s) \longrightarrow P_4(g)\right]$ white is 59 KJ/mol and enthalpy of atomization is 316.25 KJ/mol of $P(g)$.

Now give the answers of following questions:

17. The average $P-P$ bond enthalpy in P_4 molecule is:
(A) $102\,KJ$ (B) $201\,KJ$
(C) $104\,KJ$ (D) $120\,KJ$

18. The $P-P$ bond enthalpy in red phosphorus joining the two tetrahedral is:
(A) $201\,KJ$ (B) $104\,KJ$
(C) $305\,KJ$ (D) $80\,KJ$

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **6 multiple choice questions** numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. If $z_0 = \frac{1+i}{2}$, then the value of the product $(1+z_0)(1+z_0^2)(1+z_0^{2^2})\ldots\ldots(1+z_0^{2^n})$ must be

   (A) $2^{2^{n+1}}$ (B) $(1+i)\left(1-\frac{1}{2^{2^n}}\right)$, if $n>1$

   (C) $\frac{5}{4}(1+i)$, if $n=1$ (D) 0

2. Let N be any four digit number say $x_1\,x_2\,x_3\,x_4$. Then maximum value of $\frac{N}{x_1+x_2+x_3+x_4}$ is equal to

   (A) 1000 (B) $\frac{1111}{4}$

   (C) 800 (D) None of these

3. $\int \frac{e^{\cot x}}{\sin^2 x}$ [2 ln cosec x + sin 2x] dx is equal to

   (A) $-2e^{\cot x}$ ln (cosec x) + c (B) $e^{\cot x}$ ln x + c

   (C) $e^{\cot x}$ (ln (cosec x) + c (D) none of these

4. Let P be any moving point on the circle $x^2 + y^2 - 2x = 1$. AB be the chord of contact of this point w.r.t the circle $x^2 + y^2 - 2x = 0$. The locus of the circumcentre of the triangle CAB, (C being centre of the circles) is

   (A) $2x^2 + 2y^2 - 4x + 1 = 0$ (B) $x^2 + y^2 - 4x + 2 = 0$

   (C) $x^2 + y^2 - 4x + 1 = 0$ (D) $2x^2 + 2y^2 - 4x + 3 = 0$

Space for rough work

5. ABCD is a quadrilateral with side lengths AB = 4, BC = 10, CD = 6 and AD = 6, and diagonal BD = 8 units. If the incircles of triangles ABD and BCD touch BD at P and Q respectively, then area of quadrilateral C_1PC_2Q (where C_1 and C_2 are incentres of triangle ABD and BCD respectively), is

(A) $3+\frac{5\sqrt{15}}{2}$ sq. units (B) 3 sq. Units

(C) $\frac{\sqrt{15}}{6}$ sq. units (D) none of these

6. $\lim_{x\to 0}\frac{\sin^{-1}x-\tan^{-1}x}{x^3}$ is equal to

(A) $\frac{1}{6}$ (B) $\frac{1}{2}$

(C) 2 (D) $-\frac{1}{2}$

Multiple Correct Choice Type

This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

7. Consider the equation $\tan(5\pi\cos\theta)=\cot(5\pi\sin\theta)$ for $\theta\in(0,2\pi)$, then

(A) number of real solution is 28 (B) number of real solution is 14

(C) $\sin\theta+\cos\theta=\frac{2n+1}{10}$ (D) $\sin\theta-\cos\theta=\frac{2n+1}{10}$

8. Three straight lines are drawn through a point 'P' lying in the interior of the ΔABC and parallel to its sides. The areas of the three resulting triangles with P as the vertex are s_1, s_2 and s_3 then area of ABC =

(A) $\left(\sqrt{s_1}+\sqrt{s_2}+\sqrt{s_3}\right)^2$ (B) $(s_1+s_2+s_3)$

(C) $\sqrt{s_1s_2+s_2s_3+s_3s_1}$ (D) $s_1+s_2+s_3+2\left(\sqrt{s_1s_2}+\sqrt{s_2s_3}+\sqrt{s_3s_1}\right)$

9. If k denotes the remainder on dividing $\sum_{r=1}^{200} r!$ by n then (k, n) can be

(A) (5, 14) (B) (1, 3)

(C) (3, 10) (D) (3, 15)

Space for rough work

10. The square roots of the matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ are

(A) $\pm\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ (B) $\pm\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

(C) $\pm\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (D) $\pm\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

11. Range of the function $f(x)=\sqrt{\frac{1-|x|}{2-|x|}}$ contains

(A) $[0,1)$ (B) $[1,2)$

(C) $(1,2)$ (D) $(2,\infty)$

12. In ΔABC, if $A=45°, B=75°$ then

(A) $a+\sqrt{2}c=2b$ (B) $\sqrt{3}a=\sqrt{2}c$

(C) $a=\left(\sqrt{3}-1\right)b$ (D) $3a=5b$

13. Equation of plane through $A(1,0,0), B(0,1,0)$ and making an angle $\frac{\pi}{4}$ with the plane $x+y=3$ can be

(A) $x-y+2z=1$ (B) $x+y+\sqrt{2}z=1$

(C) $x+y-\sqrt{2}z=1$ (D) $x-y-2z=1$

14. If the area bounded by the curve $y=x-x^2$ and the line y = mx equals $\frac{9}{2}$. Then the value of m is:

(A) -4 (B) -2

(C) 2 (D) 4

Space for rough work

Comprehension Type

This section contains **2 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Question Nos. 15 to 16

Locus of the centre of a circle touching the circle $x^2 + y^2 - 4y - 2x = \frac{7}{4}$ internally and tangents on which from (1, 2) is making an angle 60° with each other is director circle (C) of a variable ellipse *E*. On this basis given answers to the following questions

15. The equation of the director circle (C) is

(A) $(x-2)^2 + (y-1)^2 = \sqrt{3}$ (B) $(x-1)^2 + (y-2)^2 = 3$

(C) $(x+1)^2 + (y-2)^2 = \sqrt{3}$ (D) $(x-1)^2 + (y+2)^2 = 3$

16. Locus of focus of variable ellipse(*E*) when major axis is parallel to *x*–axis, is

(A) $\left\{(x,\ y)/\sqrt{\frac{3}{2}} < |\ x-1\ | < \sqrt{3} \text{ and } y = 2\right\}$ (B) $\left\{(x,\ y)/\ 0 < |\ x\ | < \sqrt{3} \text{ and } y = 2\right\}$

(C) $\left\{(x,\ y)/\ 0 < |\ x+1\ | < \sqrt{3} \text{ and } y = 2\right\}$ (D) $\left\{(x,\ y)/\ 0 < |\ x-1\ | < \sqrt{3} \text{ and } y = 2\right\}$

Space for rough work

Paragraph for Question Nos. 17 to 18

Let S = {1, 2, 3,…..n} be a set and two numbers x and y are drawn with replacement from S. Let P_k be the probability that $x^2 - y^2$ is divisible by k & [x] denotes greatest integer less than or equal to x.

17. The value of P_2 is equal to

(A) $\frac{1}{2}$

(B) $\left(\frac{\left[\frac{n}{2}\right]}{n}\right)^2$

(C) $1-2.\frac{\left[\frac{n}{2}\right]}{n}+2\left(\frac{\left[\frac{n}{2}\right]}{n}\right)^2$

(D) $1-\frac{2-\left[\frac{n}{2}\right]}{n}-2\left(\frac{\left[\frac{n}{2}\right]}{n}\right)^2$

18. The value of P_3 is equal to

(A) $\frac{\left[\frac{n}{3}\right]}{n}$

(B) $\left(\frac{\left[\frac{n}{3}\right]}{n}\right)^2$

(C) $1+\frac{\left[\frac{n}{3}\right]}{n}+\left(\frac{\left[\frac{n}{3}\right]}{n}\right)^2$

(D) $1-2.\frac{\left[\frac{n}{3}\right]}{n}+\left(\frac{\left[\frac{n}{3}\right]}{n}\right)^2$

Space for rough work